ISA CERTIFIED PERSONAL TRAINER COURSE

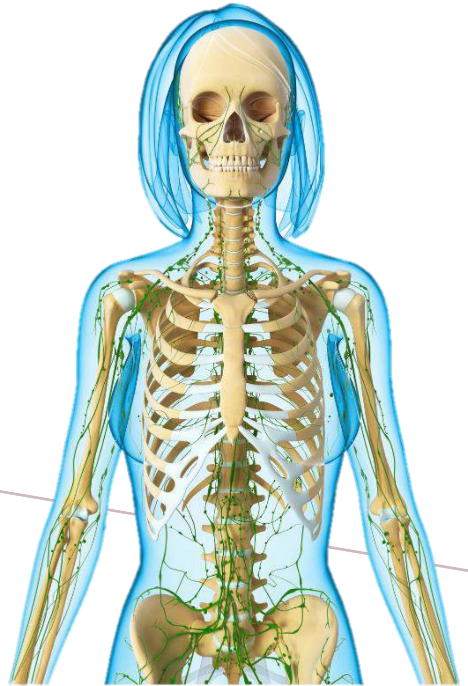
WEEK 3 – FOUNDATIONS OF SCIENCE: ANATOMY AND KINESIOLOGY

- CHAPTER 9
  MUSCULAR TRAINING:
  FOUNDATIONS AND BENEFITS

INTERNATIONAL SPORTS ACADEMY

CONTENT COVERED

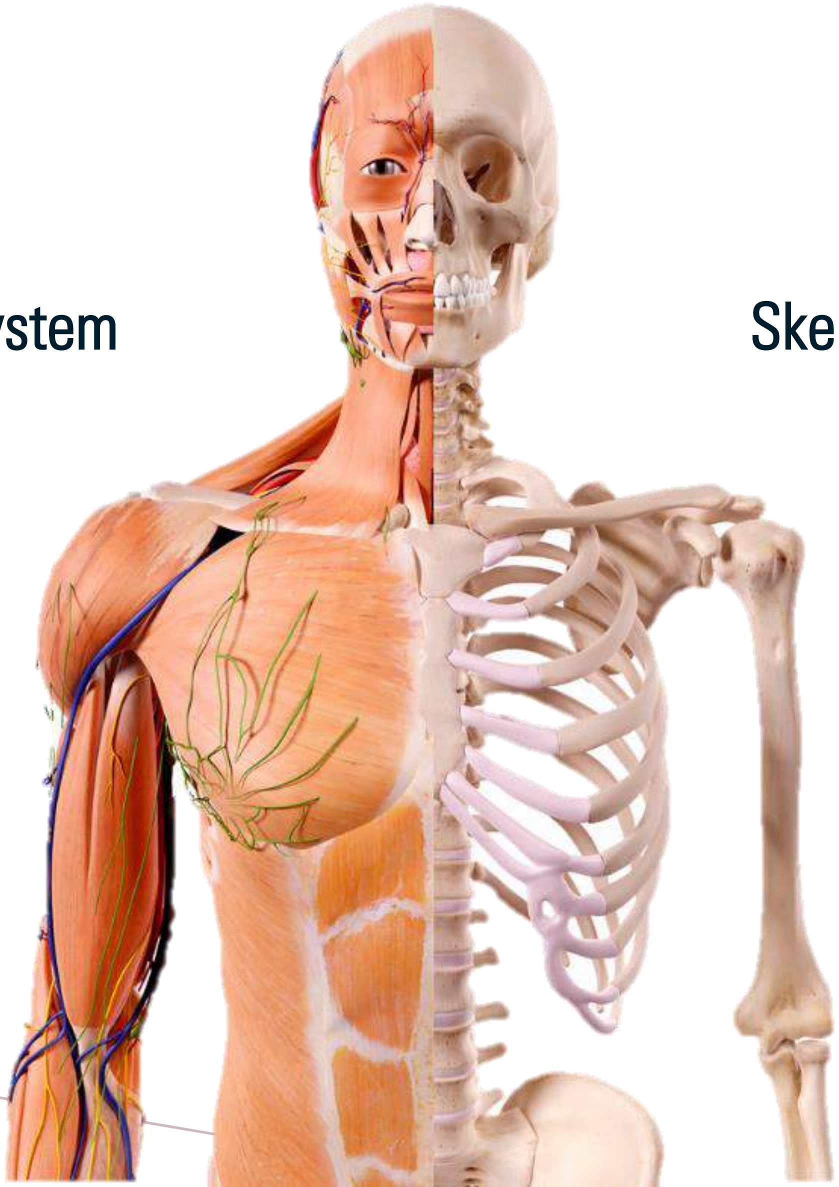
Part I: Anatomical Systems
➢ Anatomical Systems: Introduction
➢ The Skeletal System
➢ The Nervous System
➢ The Muscular System



Part II: Human Motion and Terminology
➢ Human Motion and Terminology: Introduction
➢ The Anatomical Position
➢ Fundamental Movements
➢ The 3 Planes of Motion
➢ Muscle Function and Types of Muscular Action
➢ Proprioception
➢ Kinetic Chain Movement
➢ Mobility and Stability

CHAPTER 9
MUSCULAR TRAINING: FOUNDATIONS AND BENEFITS

PART I: ANATOMICAL SYSTEMS



THE SKELETAL SYSTEM – FUNCTION AND BONES

[PG 323-326]

- Skull
- Rib Cage
- Sternum
- Vertebral Column (Cervical Spine, Thoracic Spine and Lumbar Spine)

Axial Skeleton

- Shoulder Girdle (Clavicles and Scapulae)
- Pelvic Girdle (Ilium, Ischium and Pubis)
- Upper Limbs (Humerus, Radius and Ulna)
- Lower Limbs (Femur, Tibia and Fibula)

Appendicular Skeleton

THE SKELETAL SYSTEM – MAJOR JOINTS

[PG 330]

Types of Joints

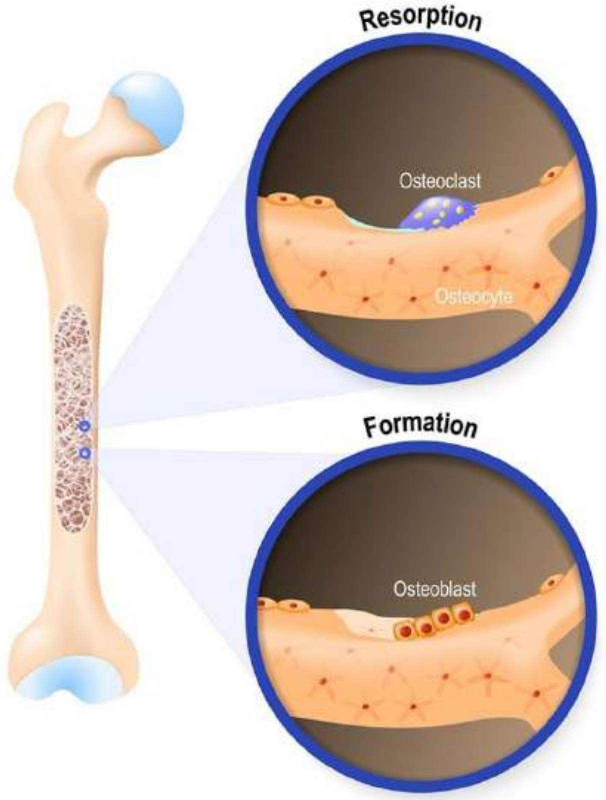
- Fibrous Joints
  - Little to no movement
- Cartilaginous Joints
  - Limited movement
- Synovial Joints
  - Freely movable
  - Most common

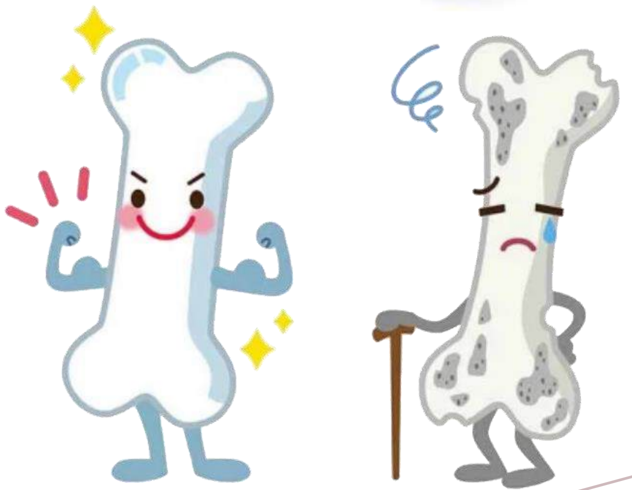
Major Synovial Joints that are typically involved in movement
➢ Elbow
➢ Scapulo-thoracic joint
➢ Gleno-humeral joint
➢ Thoracic spine
➢ Lumbar spine
➢ Hips
➢ Knees
➢ Ankles
➢ Feet

BONE REMODELLING AND WOLFF'S LAW

- Remodelling
  - Repair damage from stress / injury
  - Prevent accumulation of too much old bone
  - Involves Osteoblasts and Osteoclasts

- Wolff's Law
  - Increase in loading on a bone → bone remodelling and become stronger to resist
  - Decrease in loading on bone → bone becomes less dense and weaker



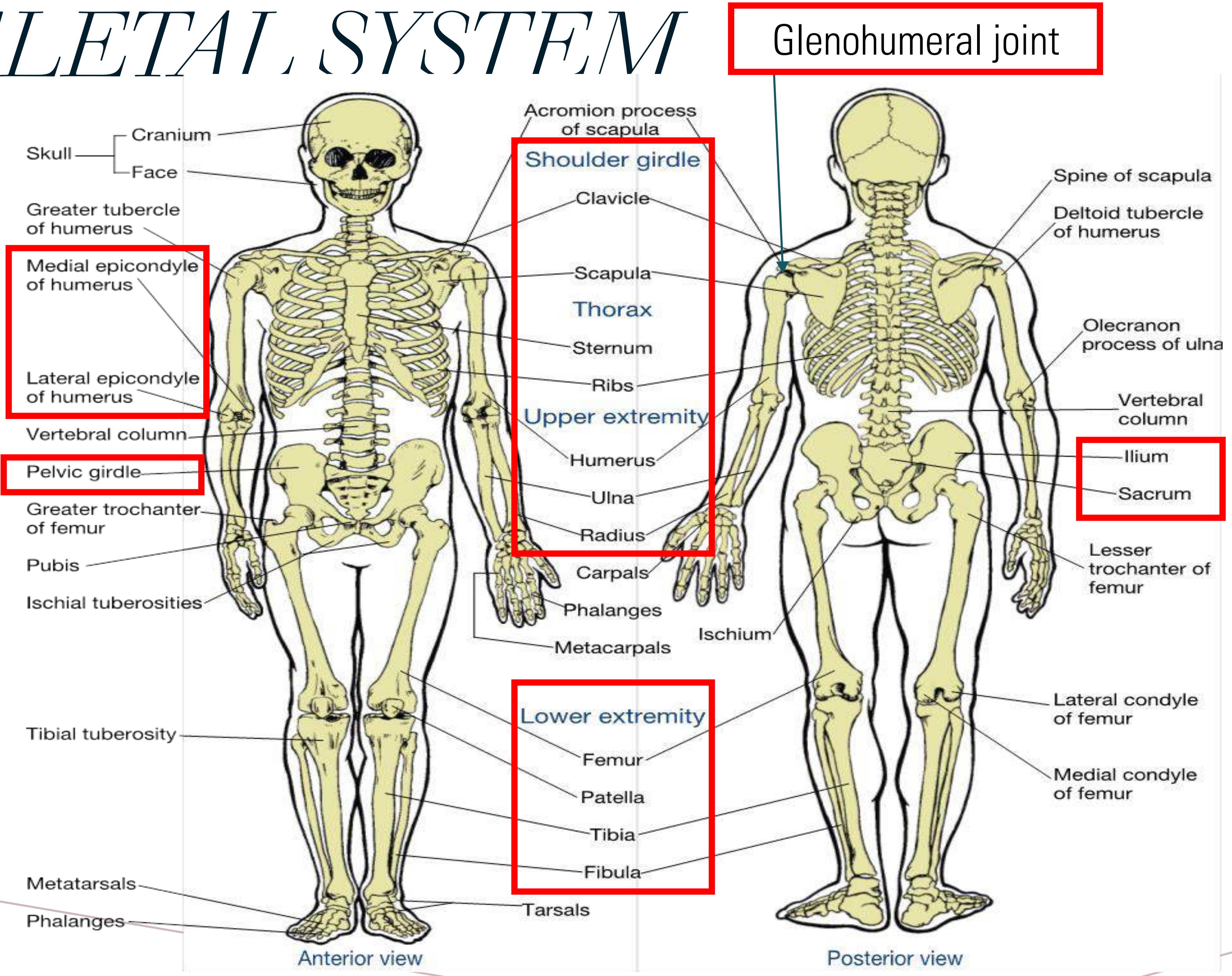


INTERNATIONAL SPORTS ACADEMY

[PG 325]

SKELETAL SYSTEM

[PG 323]

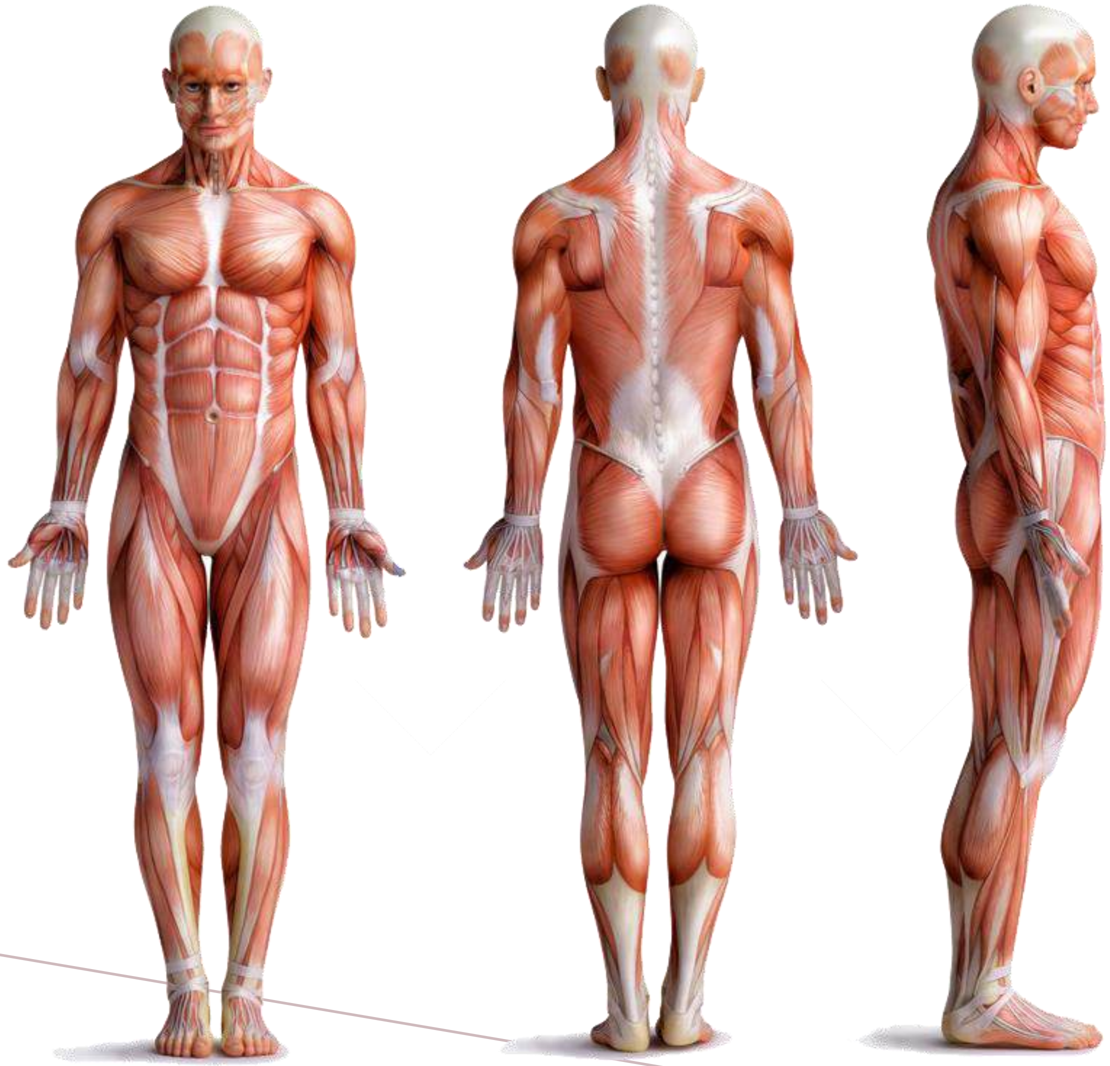


HUMAN MOTION AND TERMINOLOGY

THE ANATOMICAL POSITION

INTERNATIONAL SPORTS ACADEMY

[PG 327]



Person standing erect with the heads, eyes, and palms facing forward.

The feet are close with the toes pointing forward and the arms hanging by the sides.

FUNDAMENTAL MOVEMENTS

[PG 329-334]

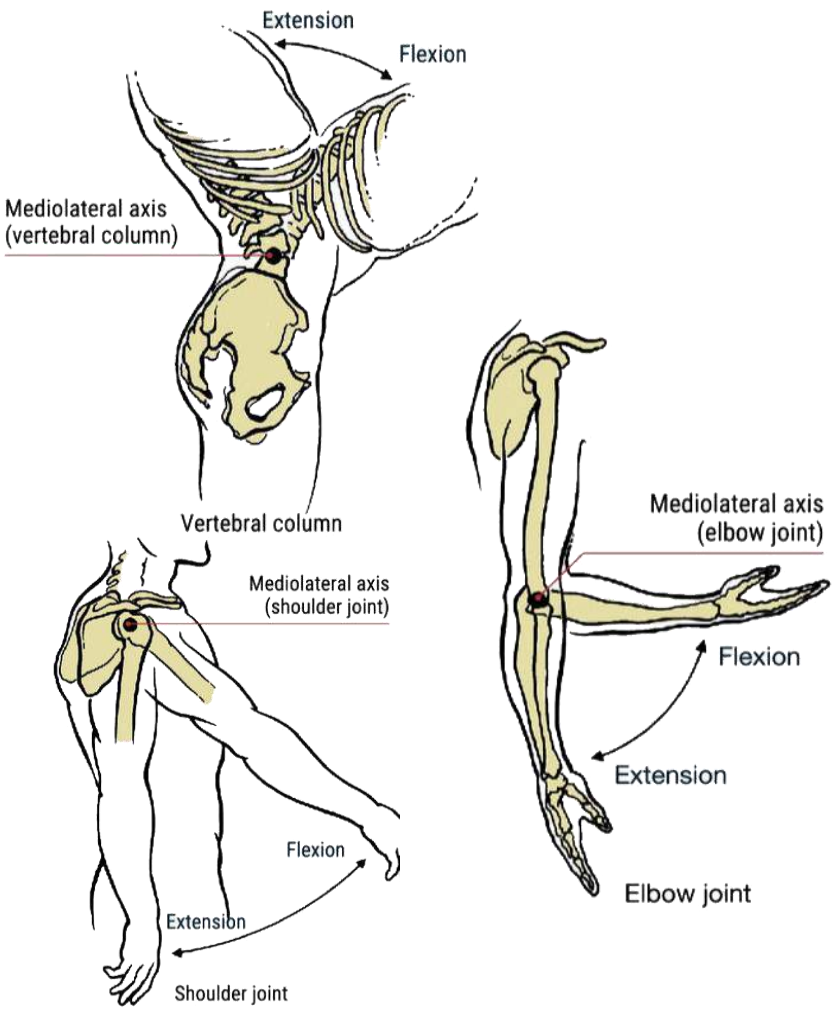


Figure 9-7: Segmental Movement in the Sagittal Plane

FLEXION

- Bones of certain joint moving toward each other
- Decrease in angle between them

EXTENSION

- Bones of certain joint moving away from each other
- Increase in angle between them

FUNDAMENTAL MOVEMENTS

[PG 329-334]

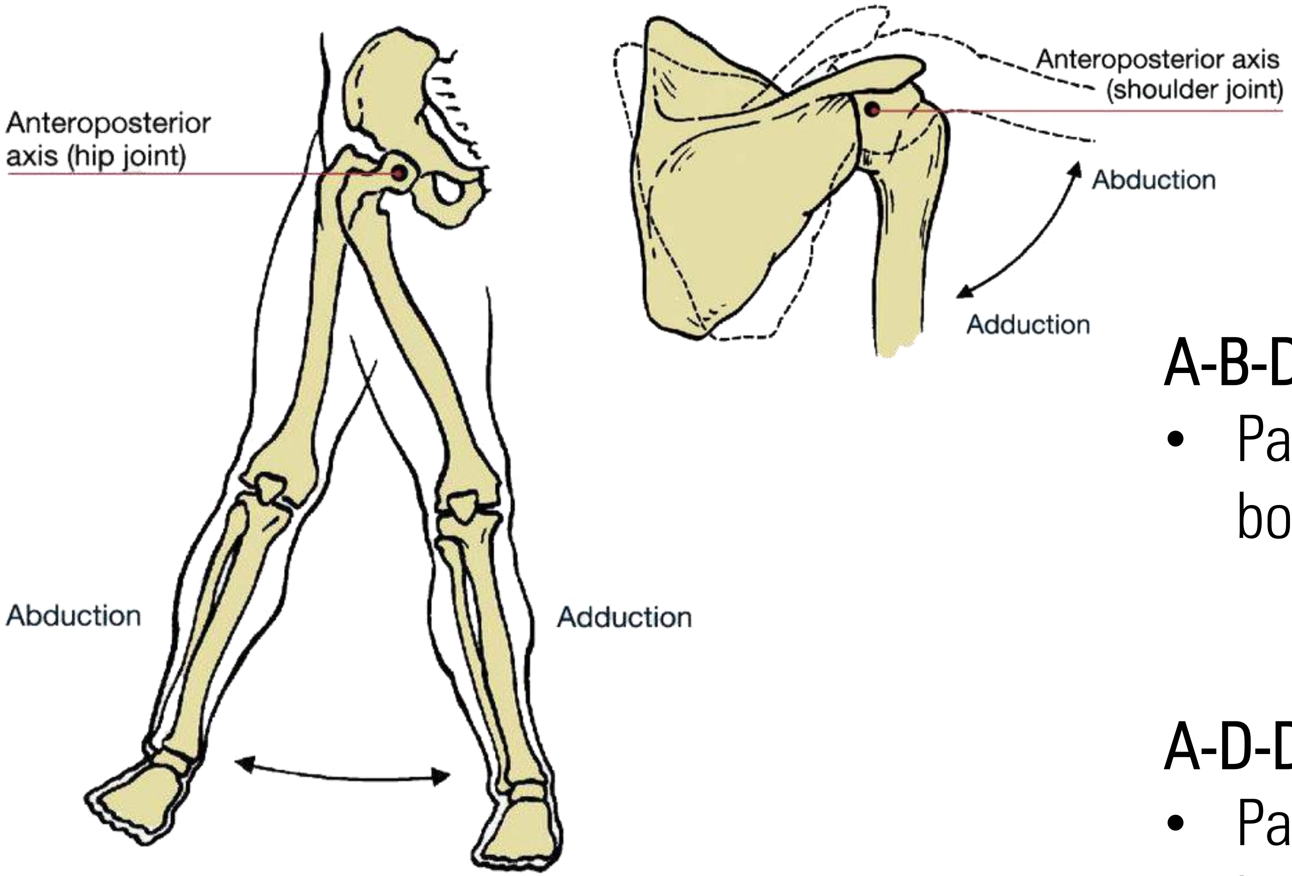


Figure 9-8: Segmental Movement in the Frontal Plane

A-B-DUCTION

- Part of body moving away from body's midline

A-D-DUCTION

- Part of body moving towards body's midline

FUNDAMENTAL MOVEMENTS

[PG 329-334]

CIRCUMDUCTION

- Flexion + Extension + Abduction + Adduction
- Certain joints like shoulders and hips are capable of such movement

FUNDAMENTAL MOVEMENTS

[PG 329-334]

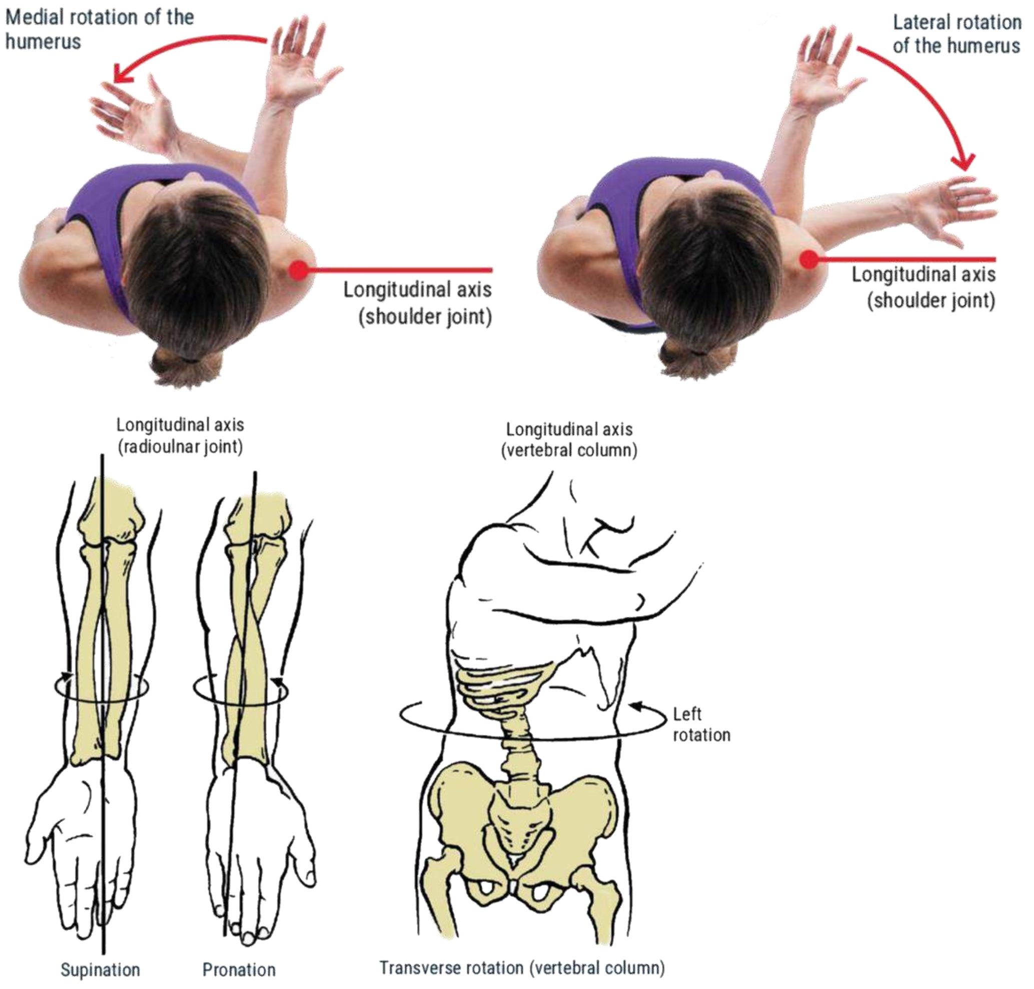


Figure 9-9: Segmental Movement in the Transverse Plane

ROTATION

- Motion of a bone around a central axis
- Shoulders, wrists and spine
- Internal and External Rotation
- Supination and Pronation

FUNDAMENTAL MOVEMENTS

INTERNATIONAL SPORTS ACADEMY

[PG 329-334]

Flexion and Extension	Plantarflexion and Dorsiflexion
Inversion / Supination and Eversion / Pronation	Abduction and Adduction
External and Internal Rotation	Elevation and depression
Horizontal Flexion / Adduction and Horizontal Extension / Abduction	
Circumduction / Opposition	

THE 3 PLANES OF MOTION

[PG 328]

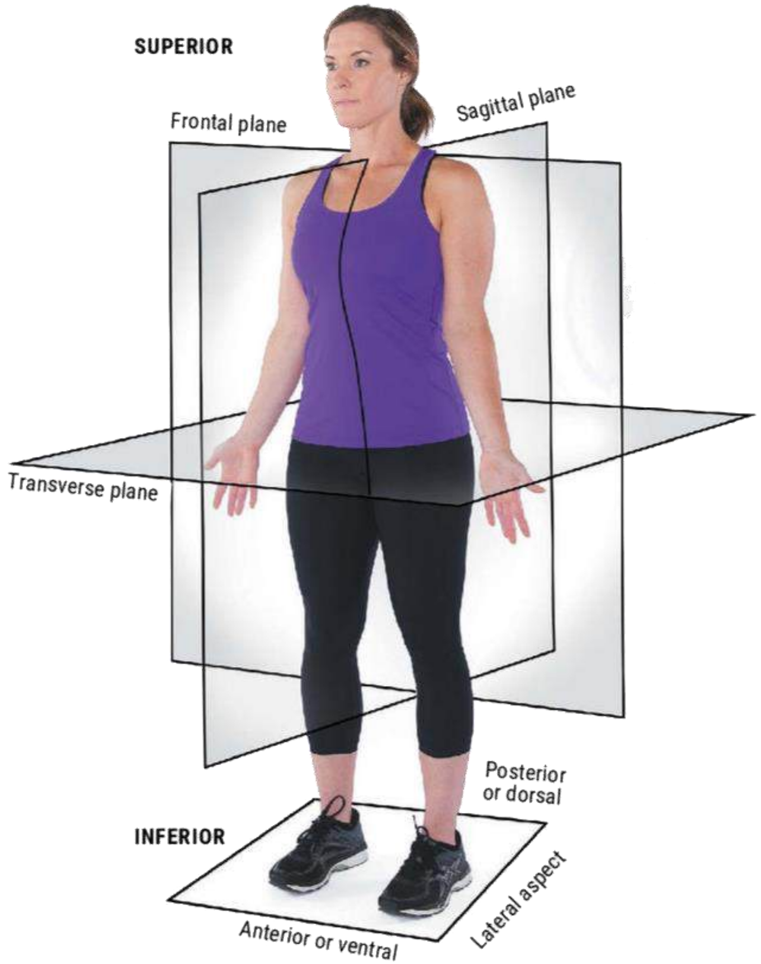


Figure 9-5: Anatomical position and planes of motion

SAGITTAL plane

- Left and Right halves
- Forward and backward movement

FRONTAL plane

- Front and Back halves
- Lateral and side movement

TRANSVERSE plane

- Top and Bottom halves
- Rotational movement

TERMINOLOGY

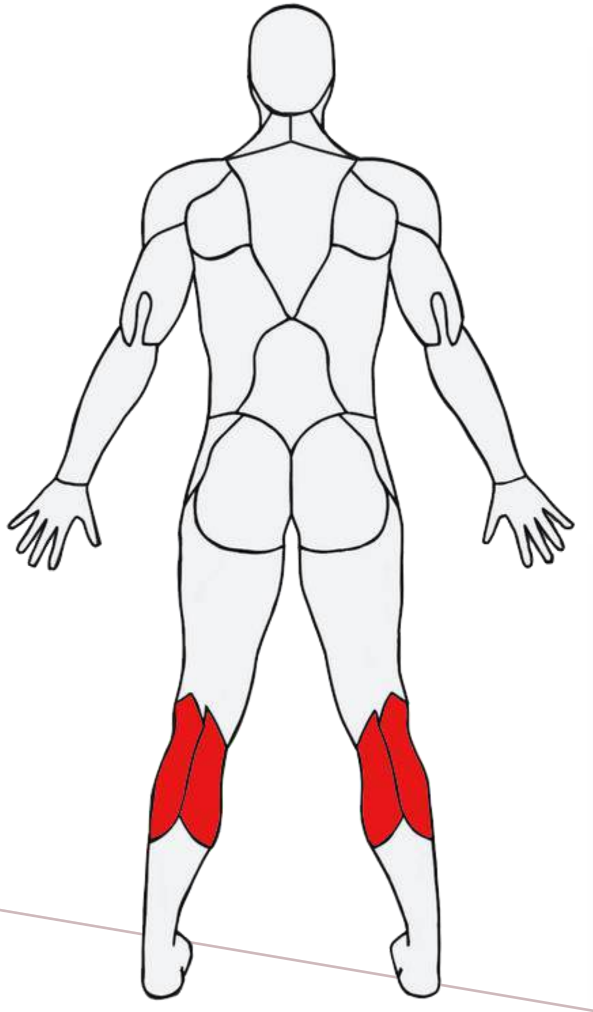
Anatomical, Directional, and Regional terms	
Anterior (ventral)	Toward the front
Posterior (dorsal)	Toward the back
Superior	Toward the head
Inferior	Away from the head
Medial	Toward the midline of the body
Lateral	Away from the midline of the body
Proximal	Toward the attached end of the limb, origin of the structure, or midline of the body
Distal	Away from attached end of the limb, origin of the structure, or midline of the body
Superficial	External; located close to or on the body surface
Deep	Internal; located further beneath the body surface than the superficial structures
Cervical	Regional term referring to the neck
Thoracic	Regional term referring to the portion of the boy between the beck and the abdomen; also known as the chest (thorax)
Lumbar	Regional term referring to the portion of the back between the abdomen and the pelvis
Plantar	The sole or the bottom of the feet
Dorsal	The top surface of the feet and hands
Palmar	The anterior or ventral surface of the hands
Sagittal plane	A longitudinal (imaginary) line that divides the body or any of its parts into right and left sections
Frontal plane	A longitudinal (imaginary) line that divides the body into anterior and posterios parts; lies at a right angle to the sagittal plane
Transverse plane	Also known as the horizontal plane; an imaginary line that divides the body or any of its parts into superior and inferior sections

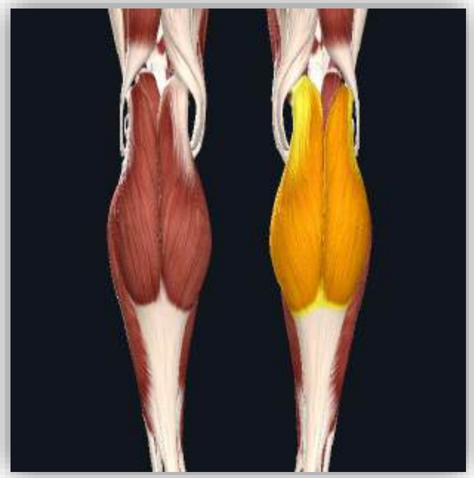
THE MUSCULAR SYSTEM

[PG 339]

MAJOR MUSCLES AND THEIR PRIMARY FUNCTIONS

[PG 341-346]





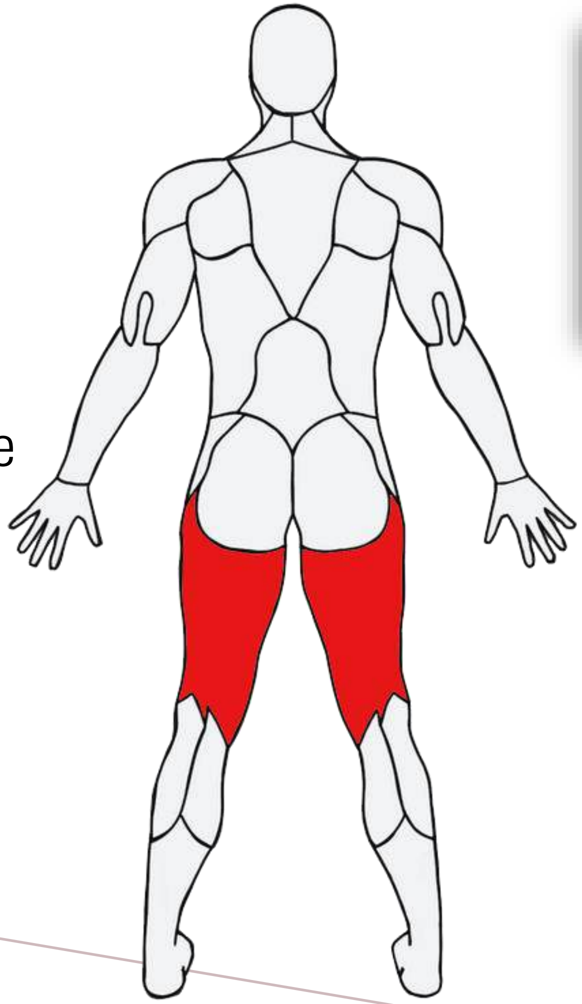
Gastrocnemius
Movement with a straight knee will target Gastrocnemius more

Soleus
Movement with bent-knee will target Soleus more

MAJOR MUSCLES AND THEIR PRIMARY FUNCTIONS

[PG 341-346]

- Hamstrings control movements along the knee and hip



Bicep Femoris

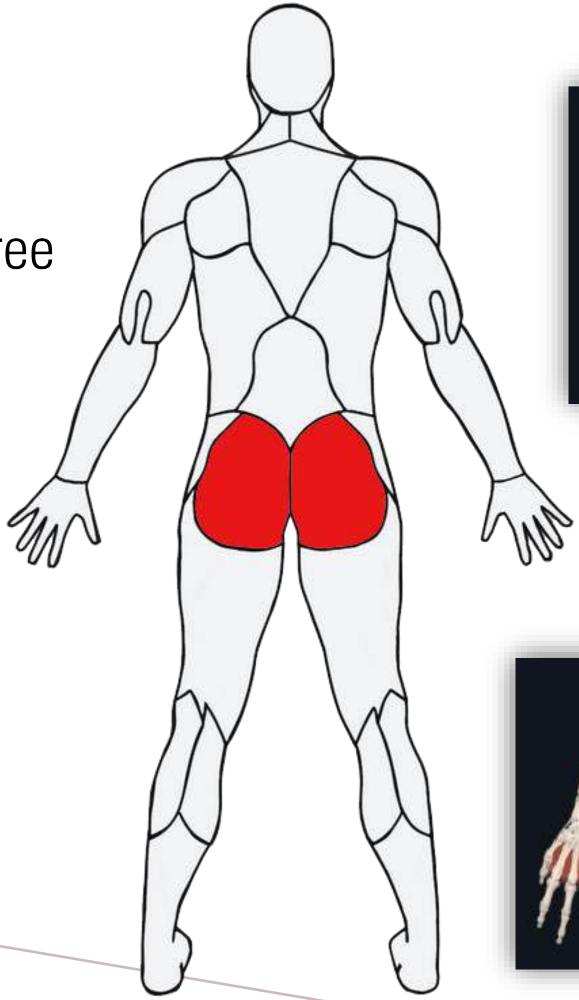
Semitendinosus

Semimembranosus

MAJOR MUSCLES AND THEIR PRIMARY FUNCTIONS

[PG 341-346]

- A group of three muscles
- Responsible for major movements along the hip joint



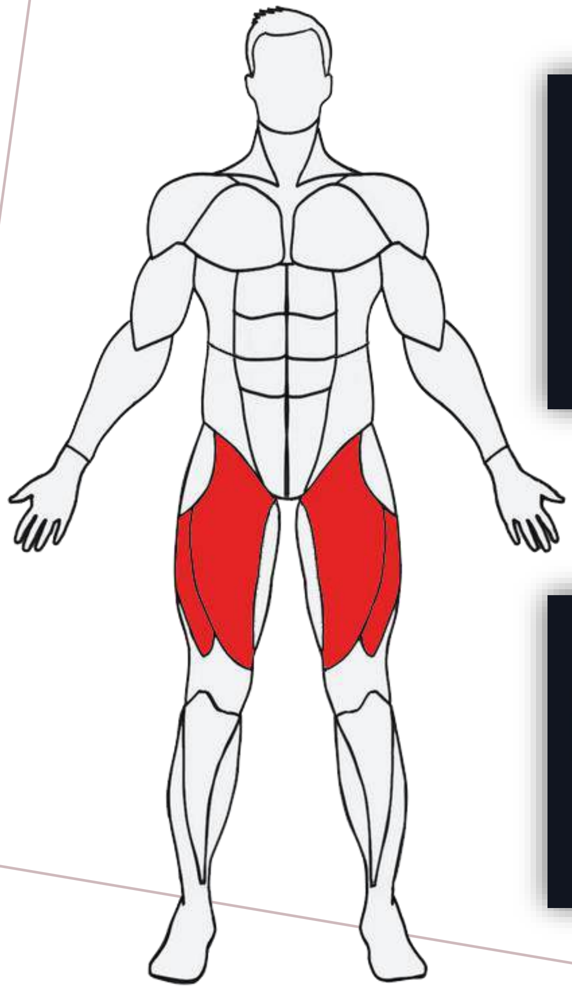
Gluteus Maximus

Gluteus Medius

Gluteus Minimus

MAJOR MUSCLES AND THEIR PRIMARY FUNCTIONS

[PG 341-346]



Rectus Femoris

Vastus Intermedius

Vastus Medialis

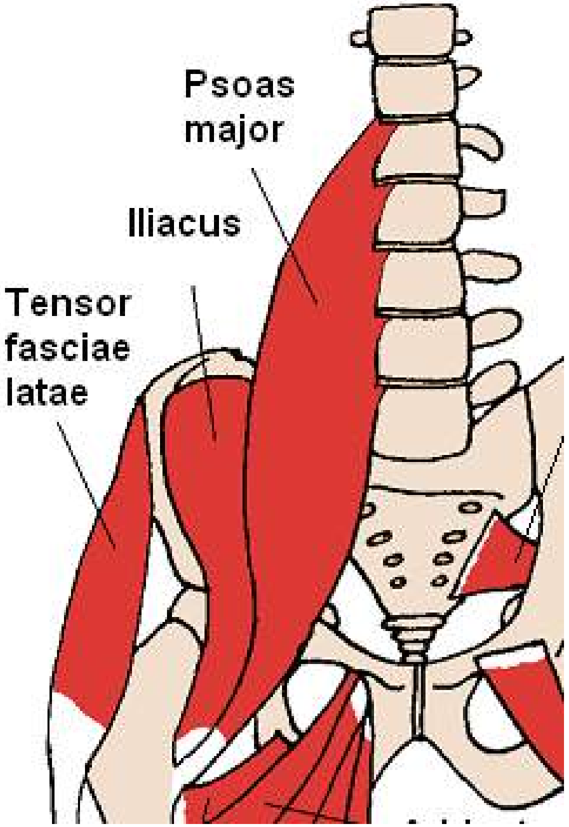
Vastus Lateralis

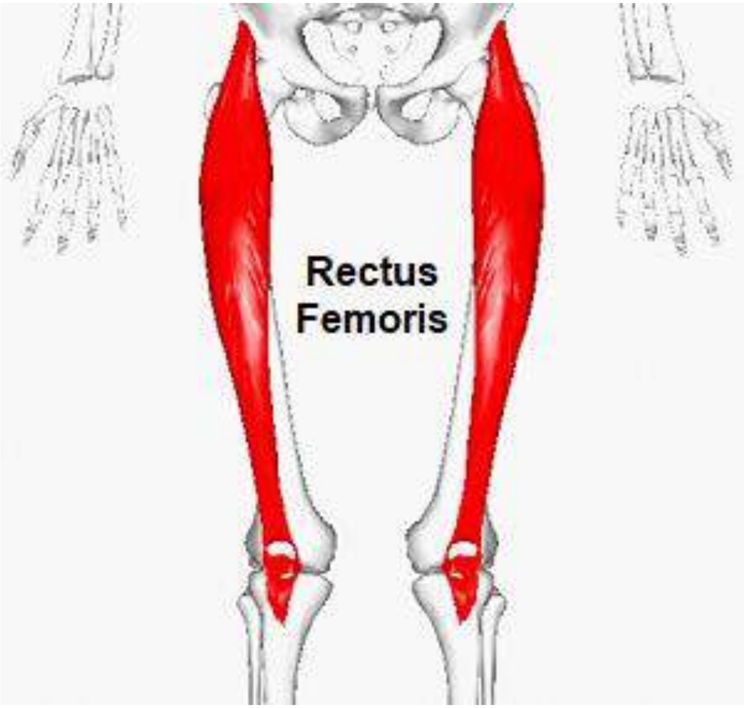
- A group of four muscles at the front
- Produce movements along the knee
- Generate a large amount of force for big movements

HIP FLEXORS

[PG 341-346]

- Iliopsoas
- Rectus femoris



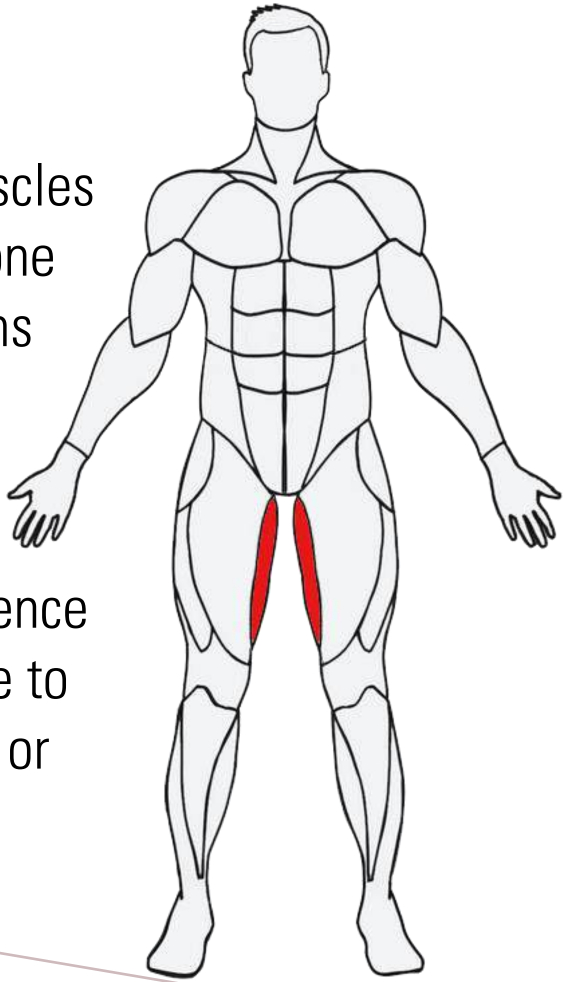


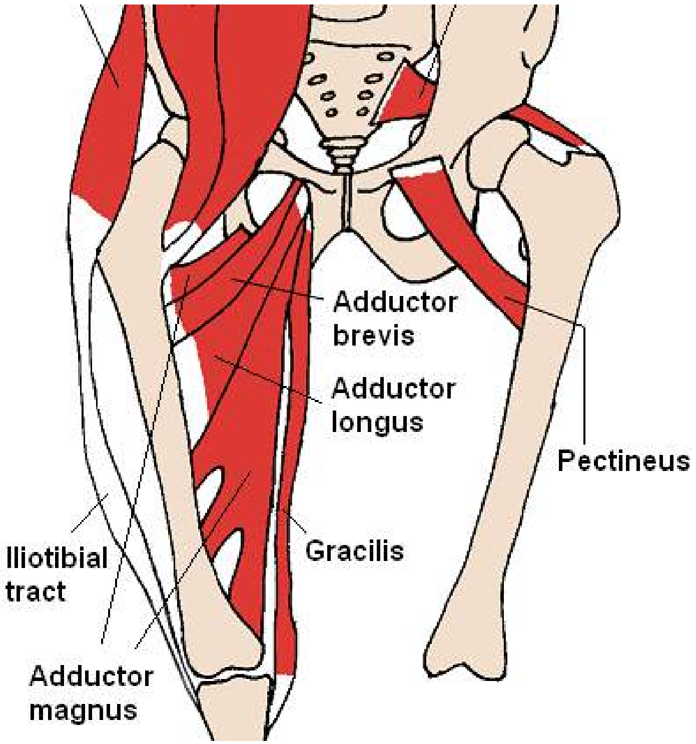
MAJOR MUSCLES AND THEIR PRIMARY FUNCTIONS

[PG 341-346]

- Group of muscles that allows one to bring thighs together

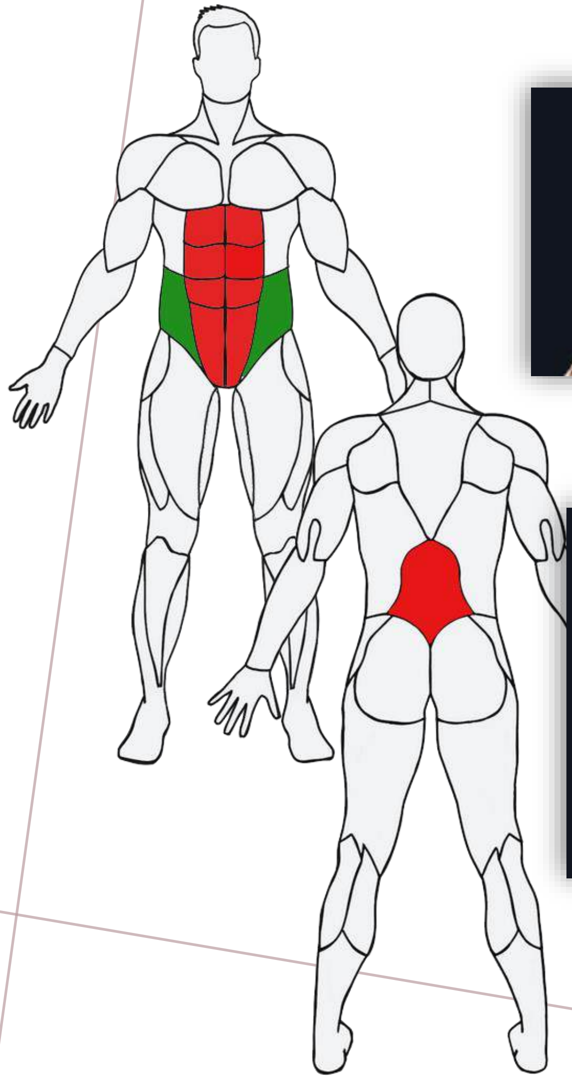
- Often experience tightness due to poor posture or inadequate stretching



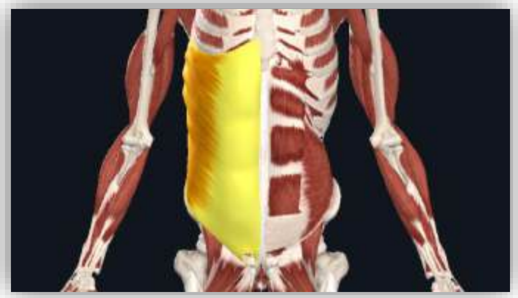


MAJOR MUSCLES AND THEIR PRIMARY FUNCTIONS

[PG 341-346]

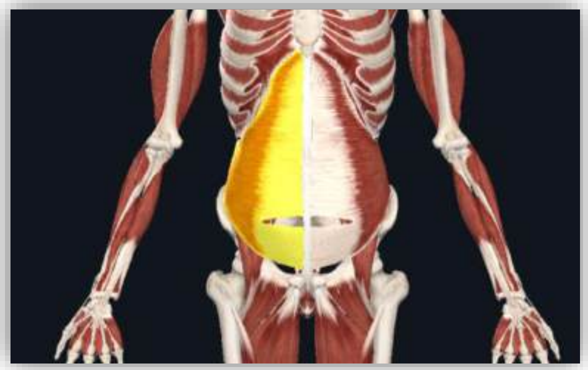


Rectus Abdominis



External & Internal Obliques

Erector Spinae



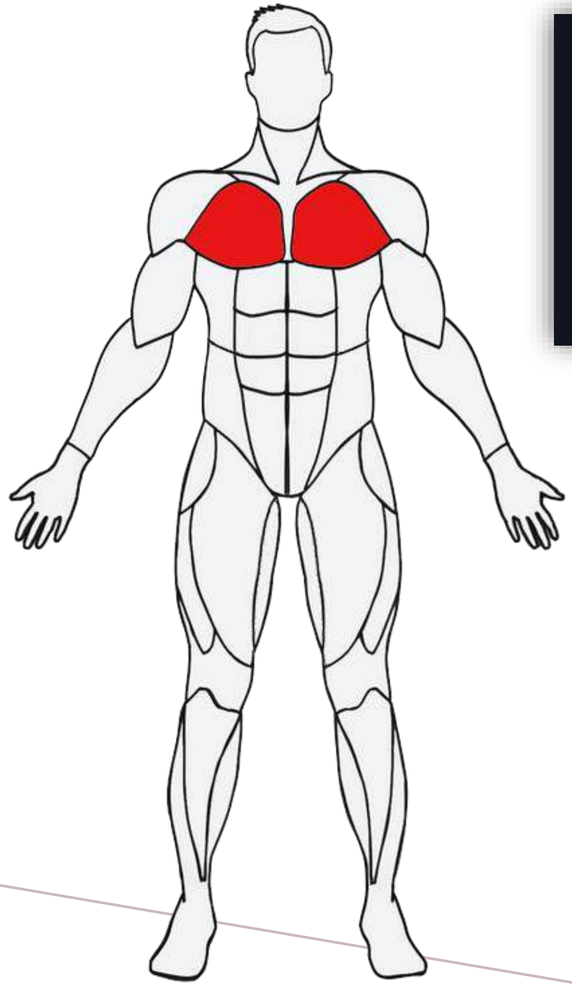
Transverse Abdominis

Quadratus Lumborum

MAJOR MUSCLES AND THEIR PRIMARY FUNCTIONS

INTERNATIONAL SPORTS ACADEMY

[PG 341-346]



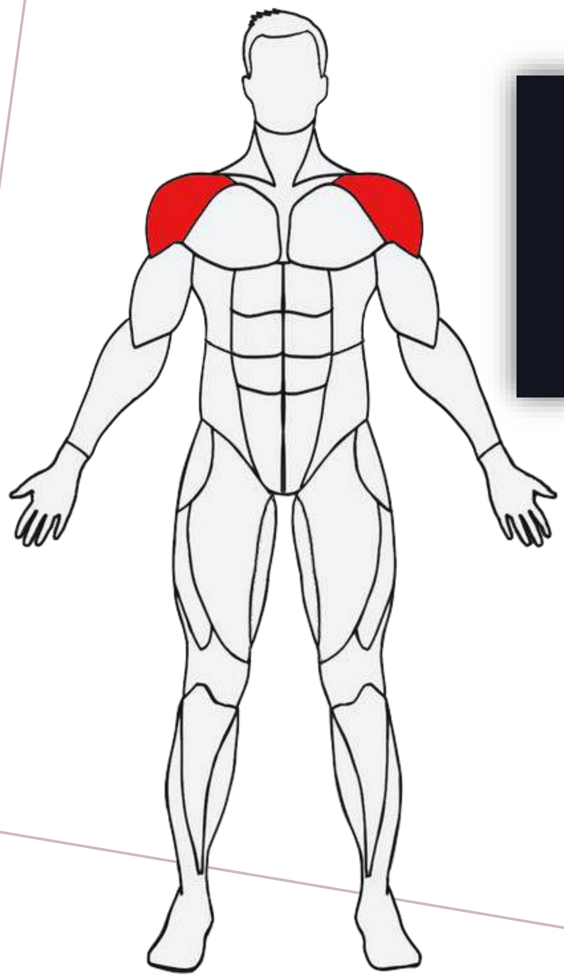
Pectoralis Major

Pectoralis Minor

- Connect the front of the chest with the bones of the upper arm and shoulder
- Allow movement of humerus and produce various movements around the shoulder joint

MAJOR MUSCLES AND THEIR PRIMARY FUNCTIONS

[PG 341-346]



Anterior Deltoid

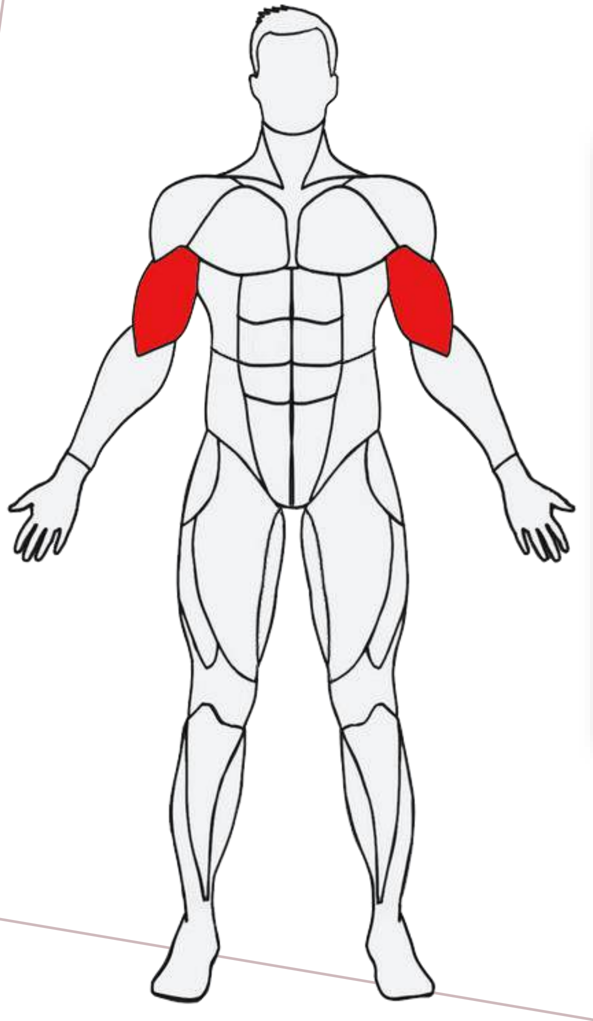
- Group of 3 muscles that make up the muscles of the Shoulder
- Widest at the top and narrows to apex as it travels down the arm

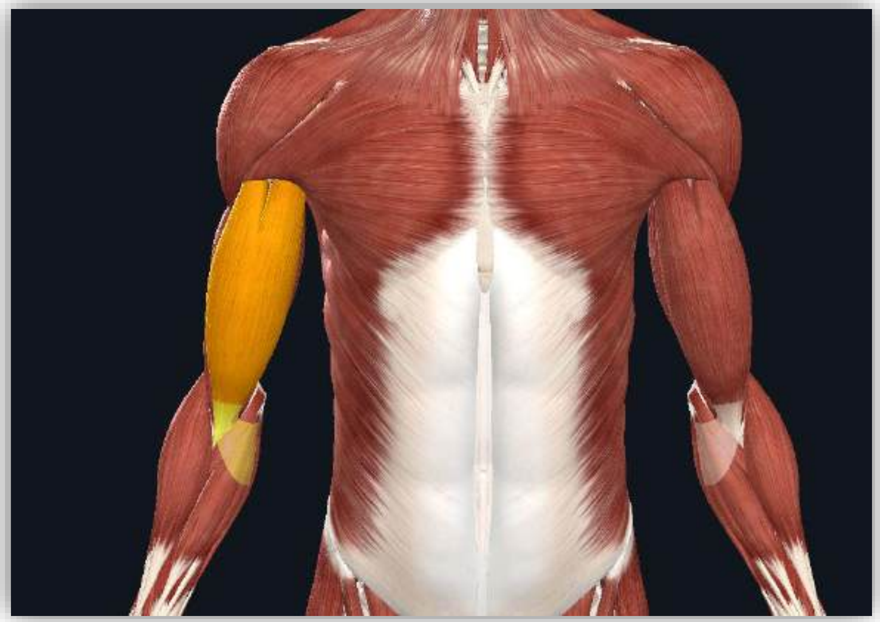
Medial Deltoid

Posterior Deltoid

MAJOR MUSCLES AND THEIR PRIMARY FUNCTIONS

[PG 341-346]



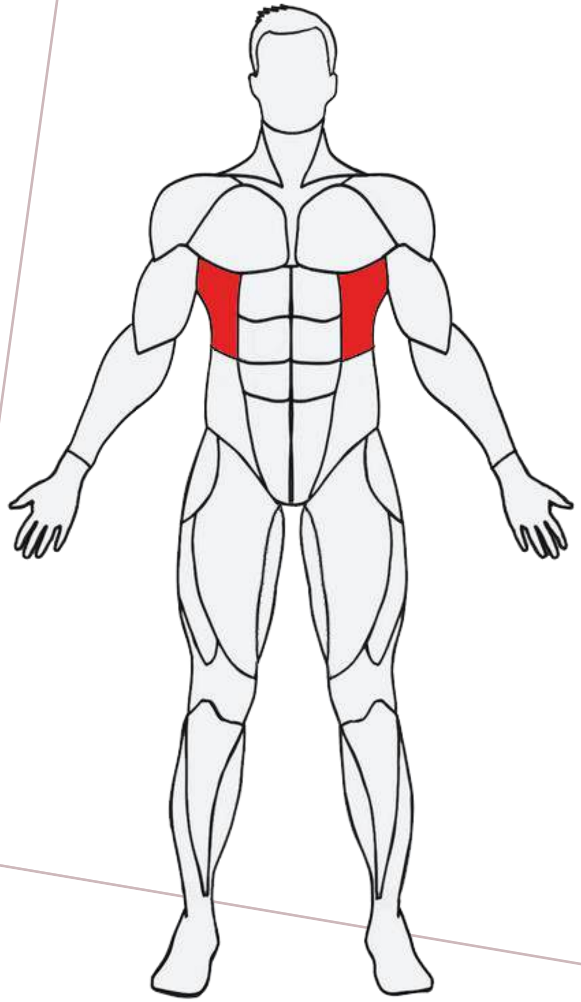


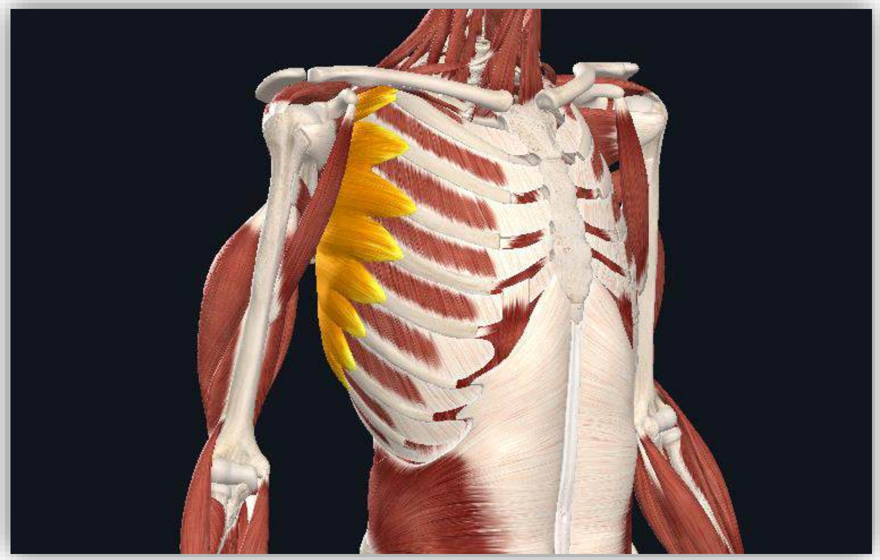
Biceps Brachii

- Muscle lies on the front of upper arm between shoulder and the elbow
- Produce movements along the elbow and the forearm

MAJOR MUSCLES AND THEIR PRIMARY FUNCTIONS

[PG 341-346]



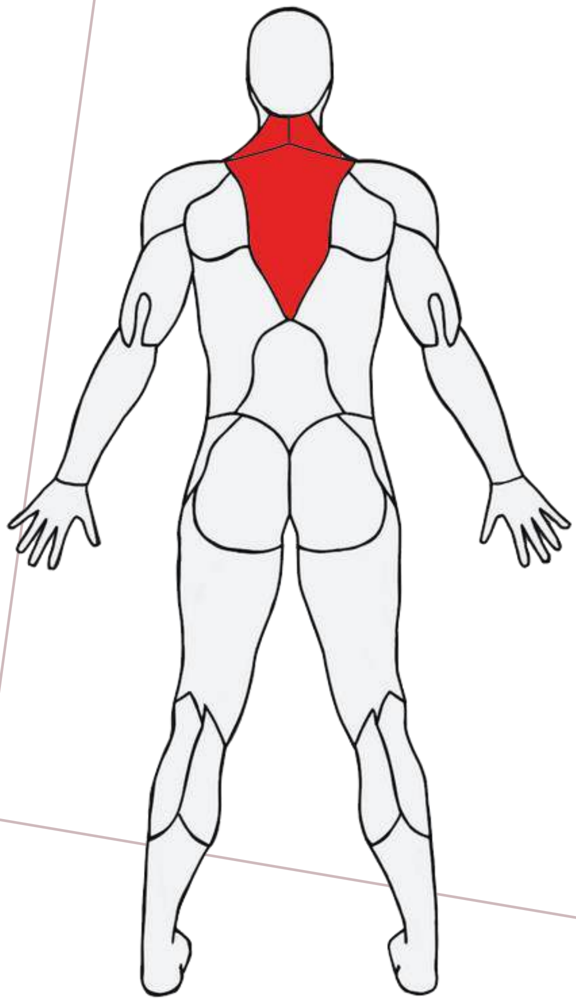


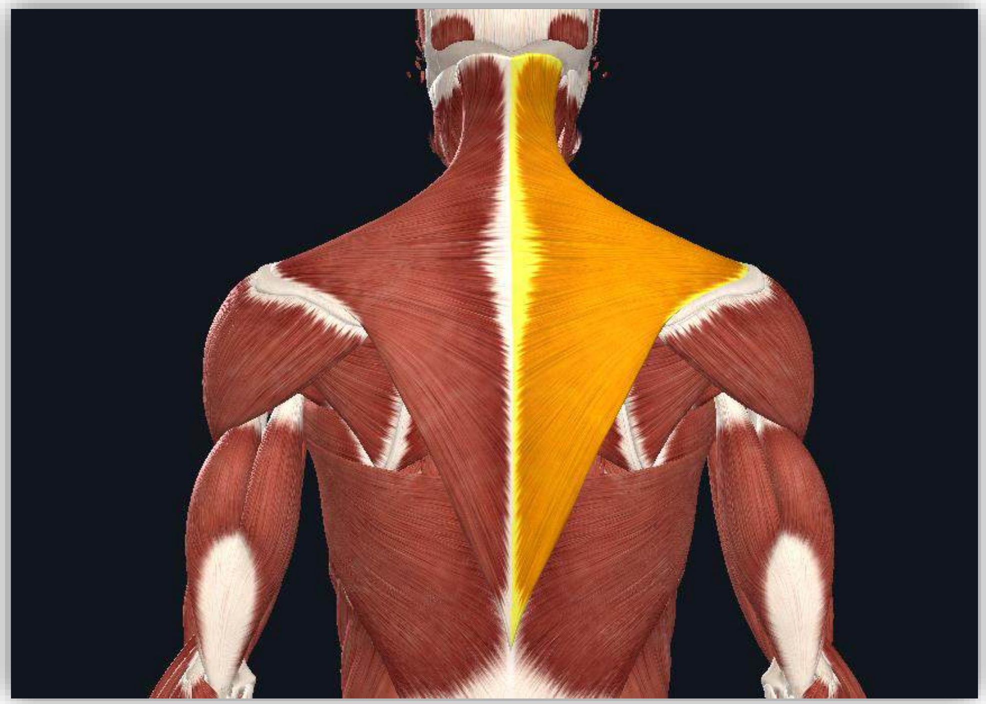
Serratus Anterior

- Pull scapulae forward around rib cage
- Responsible for anchoring scapulae onto axial skeleton
- Trained to improve scapulae stability and prevent postural deviations

MAJOR MUSCLES AND THEIR PRIMARY FUNCTIONS

[PG 341-346]





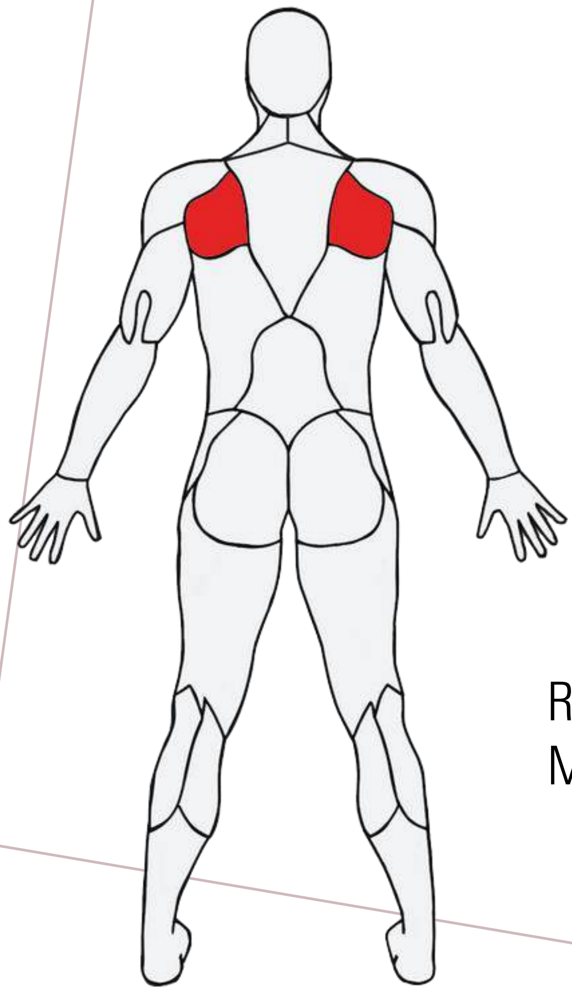
Trapezius

- Widest back muscle covering upper back and neck regions
- Produce movements along scapulae and maintain posture where head and upper back are concerned

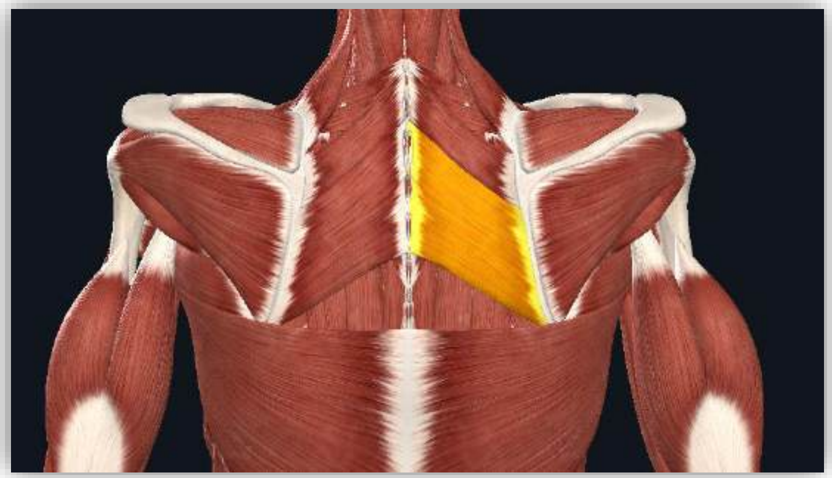
MAJOR MUSCLES AND THEIR PRIMARY FUNCTIONS

INTERNATIONAL SPORTS ACADEMY

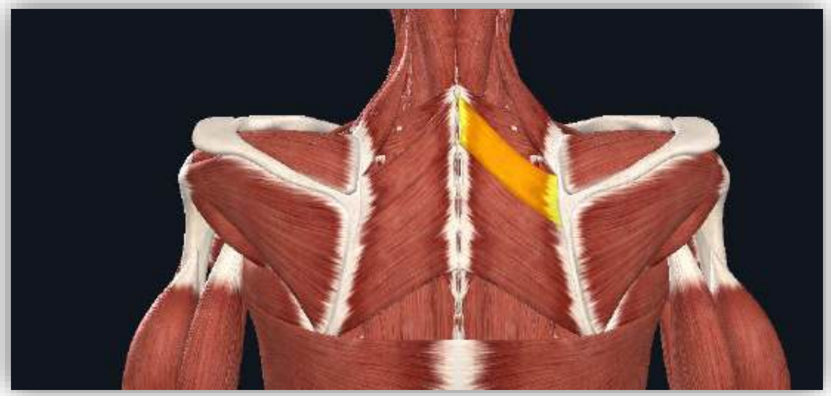
[PG 341-346]



Rhomboid Major



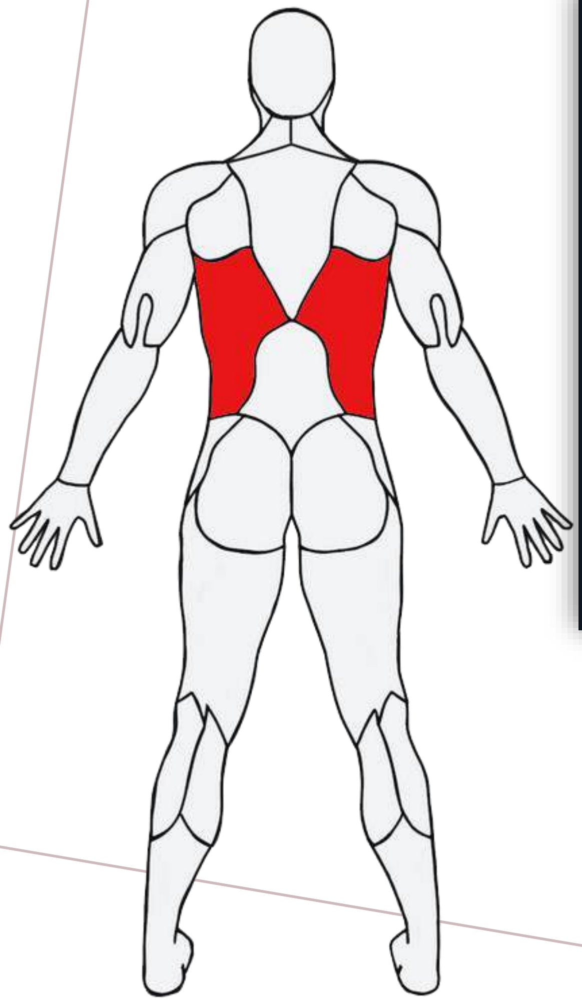
Rhomboid Minor

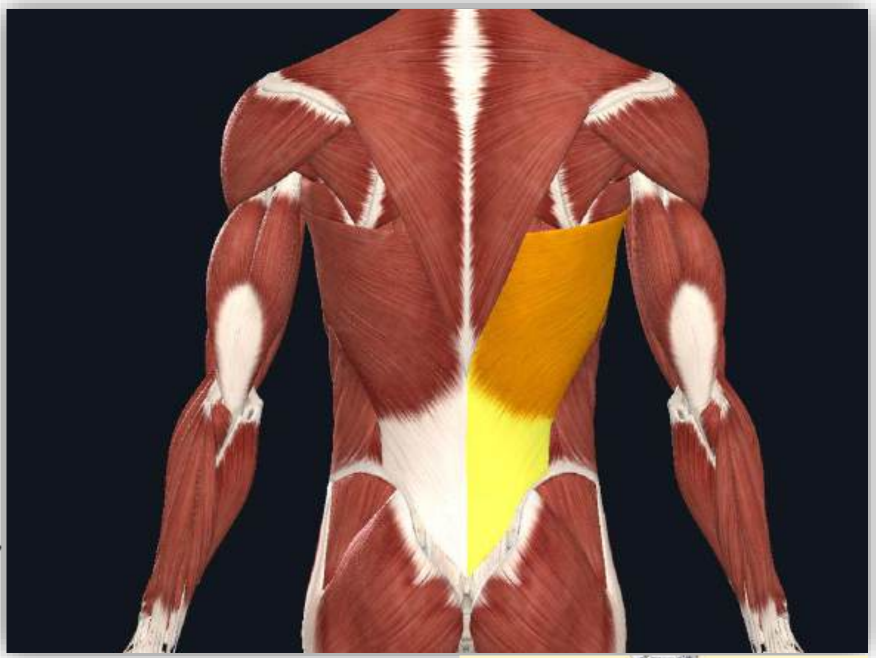


- On upper back, but underneath trapezius
- Pull shoulder blades together and rotate scapulae in a downward direction
- Stability for shoulders

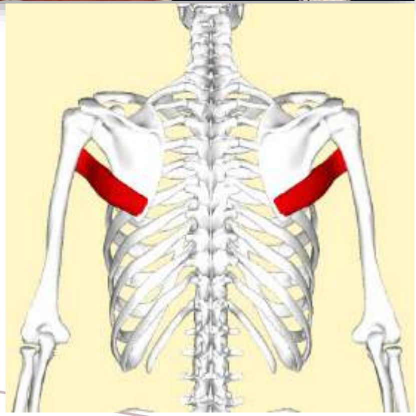
MAJOR MUSCLES AND THEIR PRIMARY FUNCTIONS

[PG 341-346]





Latissimus Dorsi



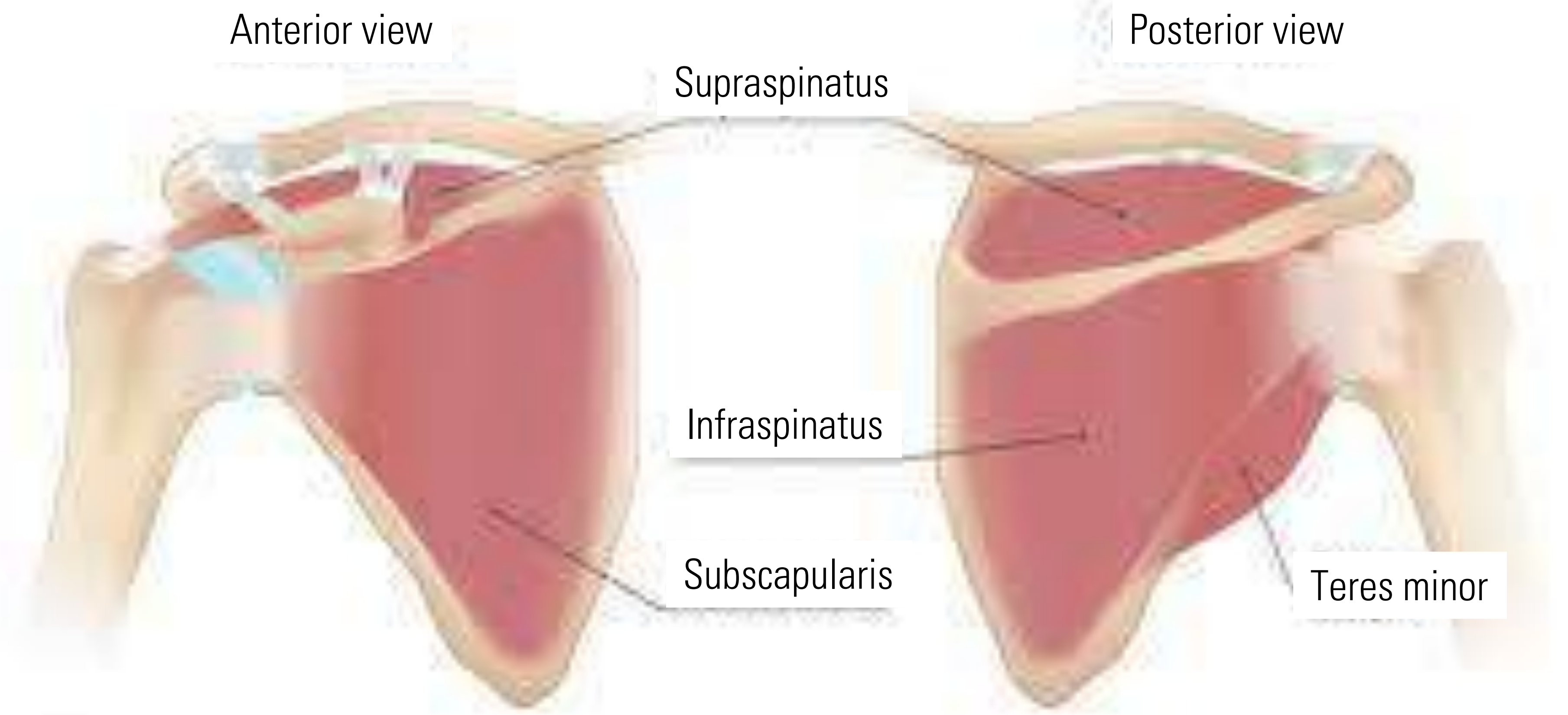
Teres Major

- Largest muscle in the upper body
- Stabilize back and produce movements along the shoulder and scapulae
- Assist with breathing process
- Teres major

MAJOR MUSCLES AND THEIR PRIMARY FUNCTIONS

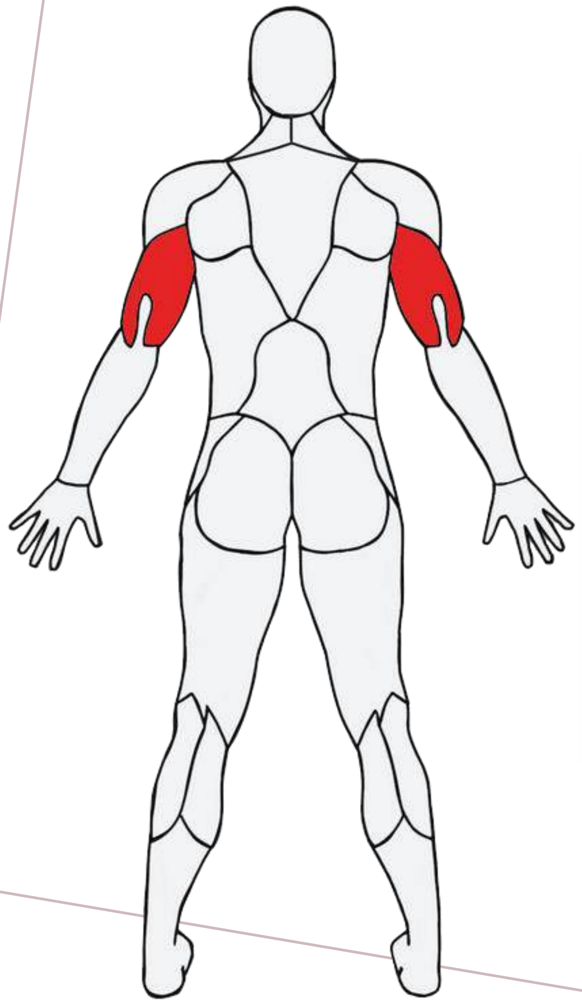
[PG 341-346]

Rotator cuff muscles



MAJOR MUSCLES AND THEIR PRIMARY FUNCTIONS

[PG 341-346]



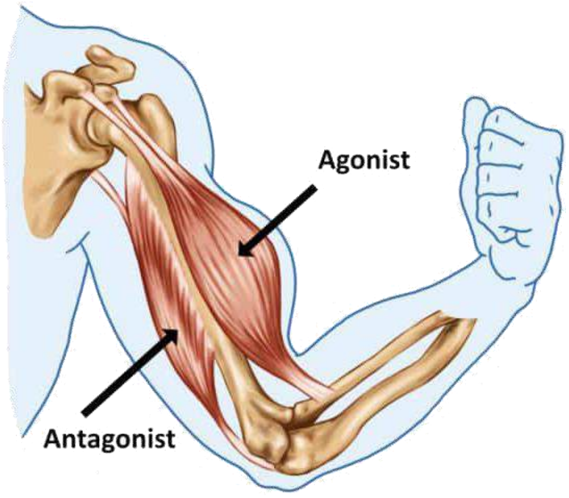
Triceps Brachii

- Almost the entire length of the humerus
- Consists of a long, medial and lateral head
- Responsible for straightening of the arm

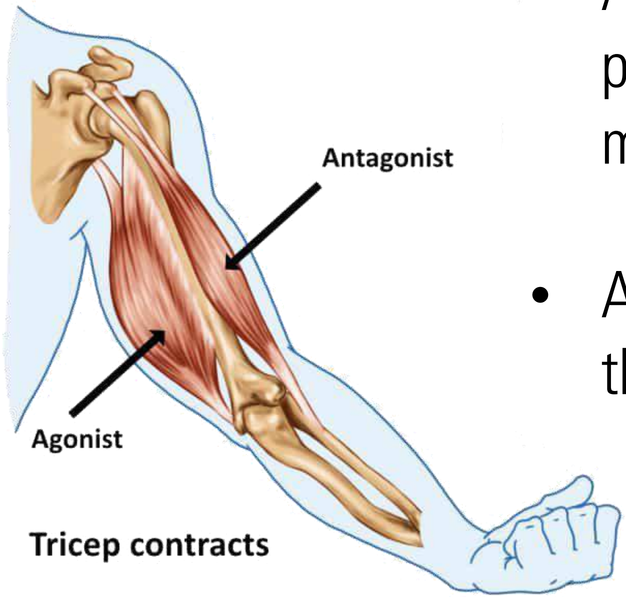
MAJOR MUSCLES AND THEIR PRIMARY FUNCTIONS

MUSCLE FUNCTION

[PG 340, 353-354]



Bicep contracts



Tricep contracts

Muscle Function

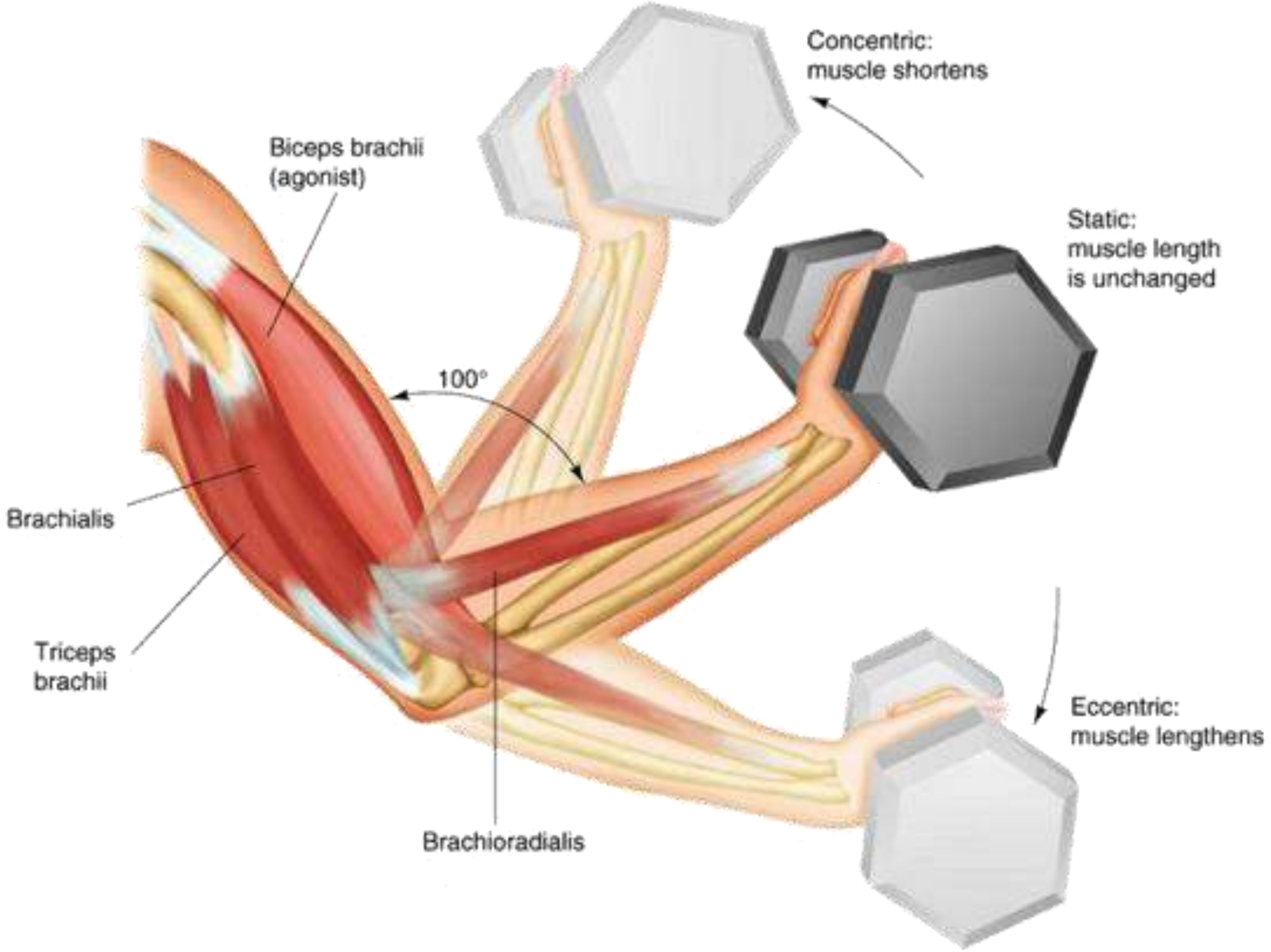
- Muscles surrounding any given joint help produce force required for movement

- Agonist → Primary muscle producing the force for a movement

- Antagonist → Opposite side of the joint

TYPES OF MUSCULAR ACTION

[PG 340, 353-354]

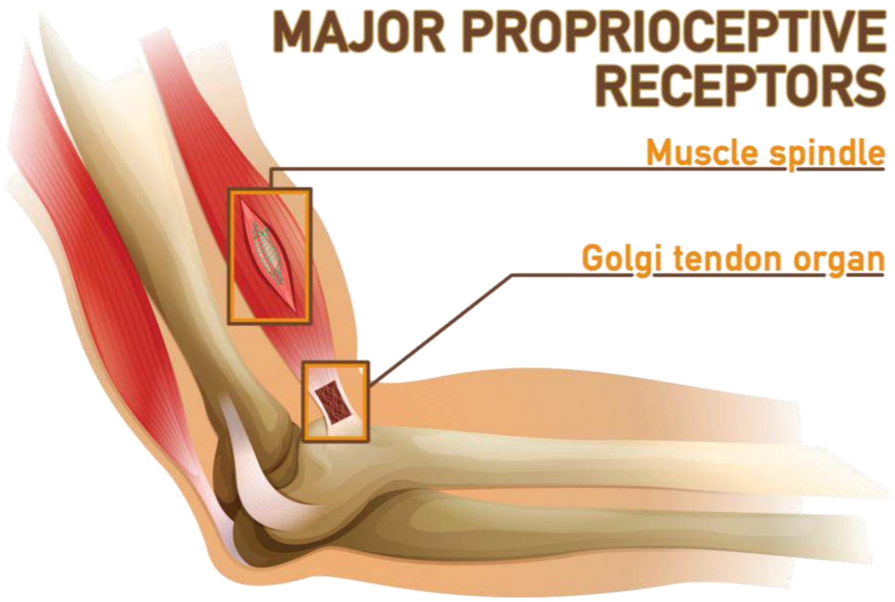


- Important to consider muscle contraction
- Actions named after the muscle's apparent length during the action
- Agonist shortens during force production, while antagonist lengthens

PROPRIOCEPTION – MUSCULOTENDINOUS RECEPTORS

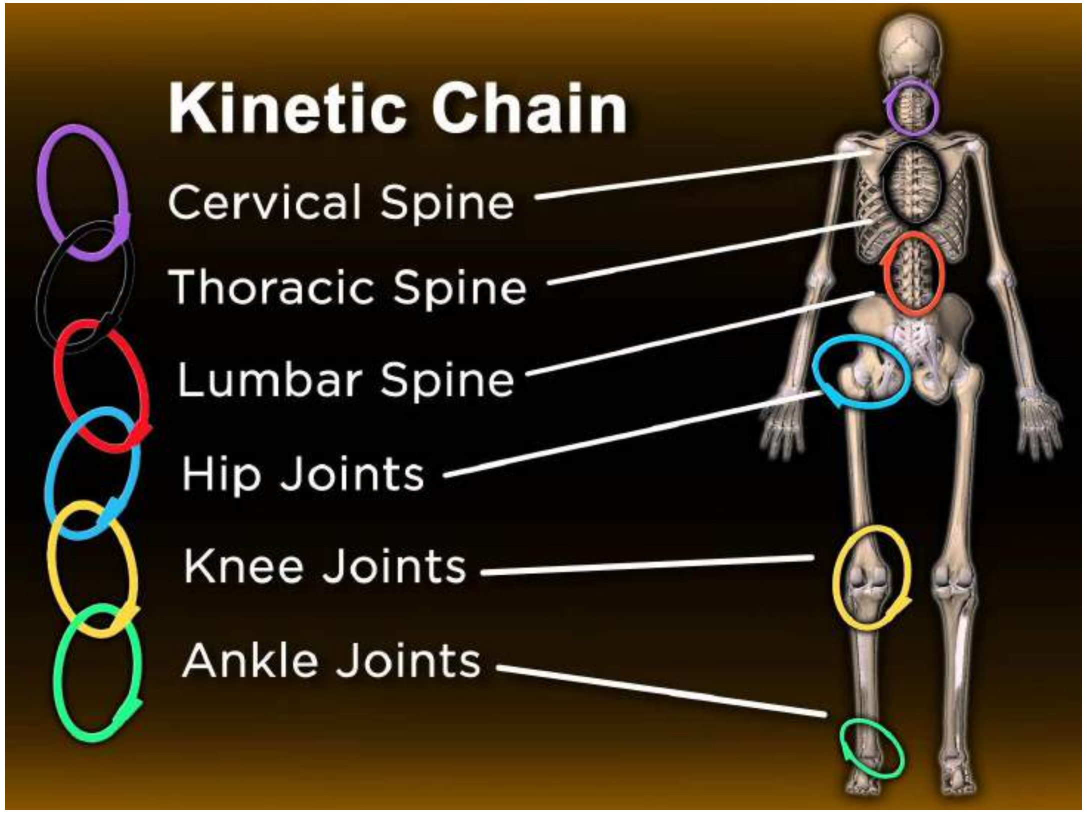
[PG 337-338]

- Proprioception
  - Sense of knowing where body is in relation to its various segments and external environment
- Proprioceptors
  - Receptors located in the skin, and around joints and muscles
- Golgi Tendon Organs (GTOs)
  - Constantly measure tension or degree of contraction and attempt to reduce any tension
  - Works with Muscle Spindles to regulate body's sense of postural control and prevent injuries



KINETIC CHAIN MOVEMENT

[PG 354-356]



- Movement along each link affects the movement of the adjacent joints
- **Open-kinetic-chain**
  - End of chain/limb farthest from body is <u>FREE</u>
- **Closed-kinetic-chain**
  - End of chain/limb farthest from body is <u>FIXED</u>

KINETIC CHAIN MOVEMENT

[PG 354-356]

Closed-Kinetic-Chain VS Open-Kinetic-Chain

MOBILITY AND STABILITY

[PG 357]

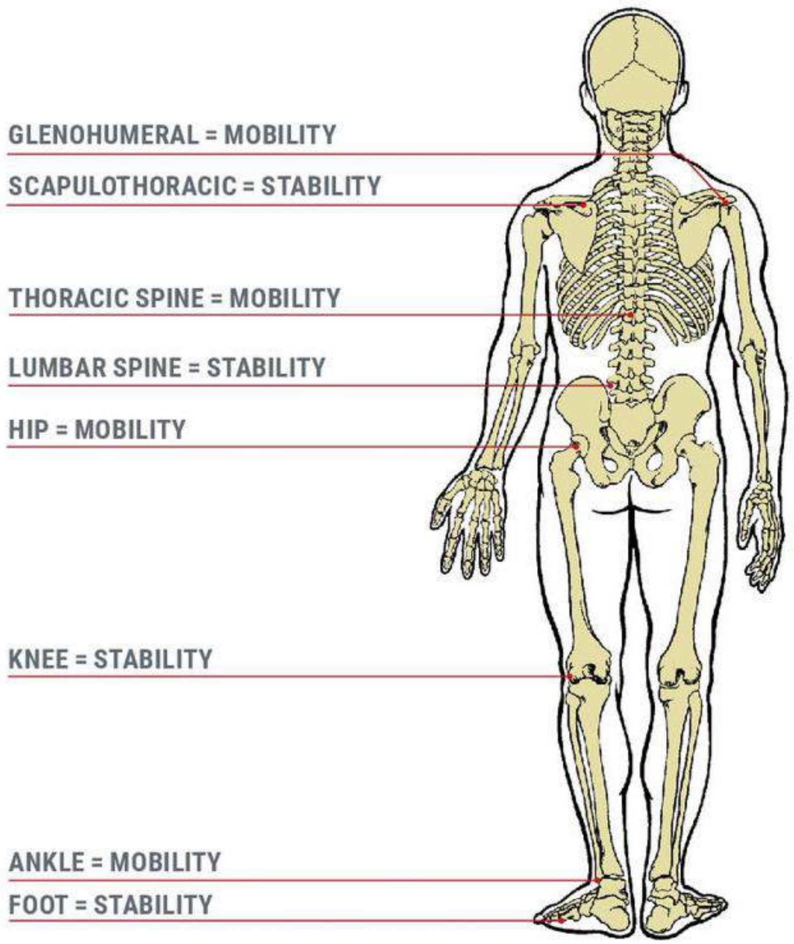
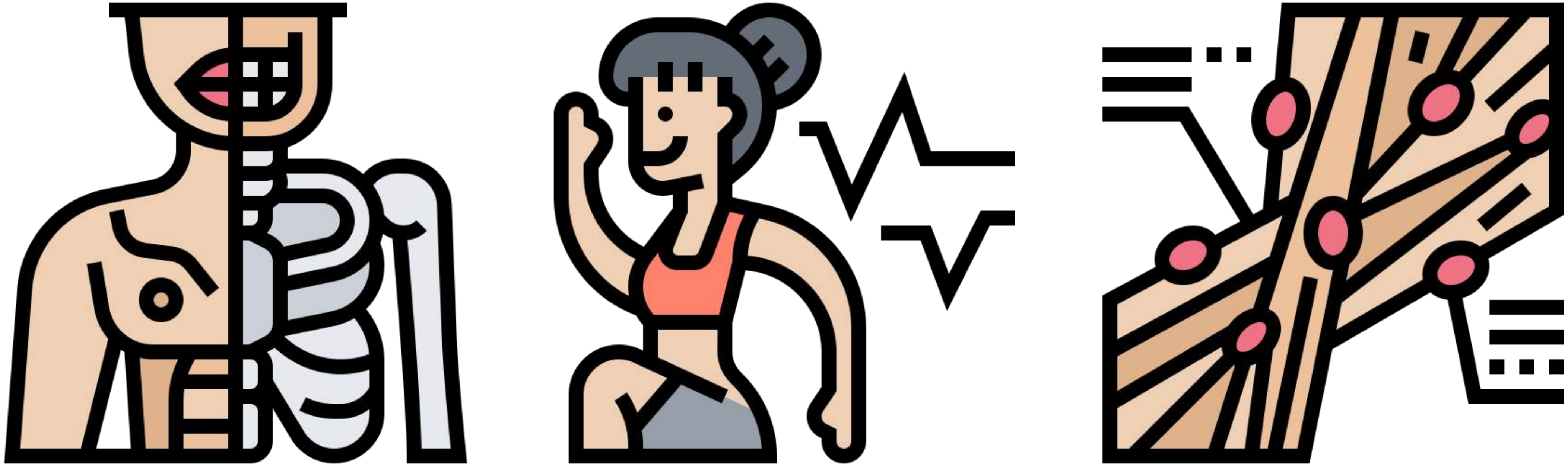


Figure 9-28: Mobility and stability of the kinetic chain

- Stability
  - Ability to control/maintain joint movement or position
- Mobility
  - Uninhibited movement around a joint or body segment
- Muscle imbalance affects mobility-stability relationships in joints → misalignments and flawed movement

LESSON CONCLUSION



Having a basic understanding of human anatomy, movement mechanics, and the muscles responsible for those movements would allow you to better design effective muscular training programs for your clients.